

BAAZ JKBOSE



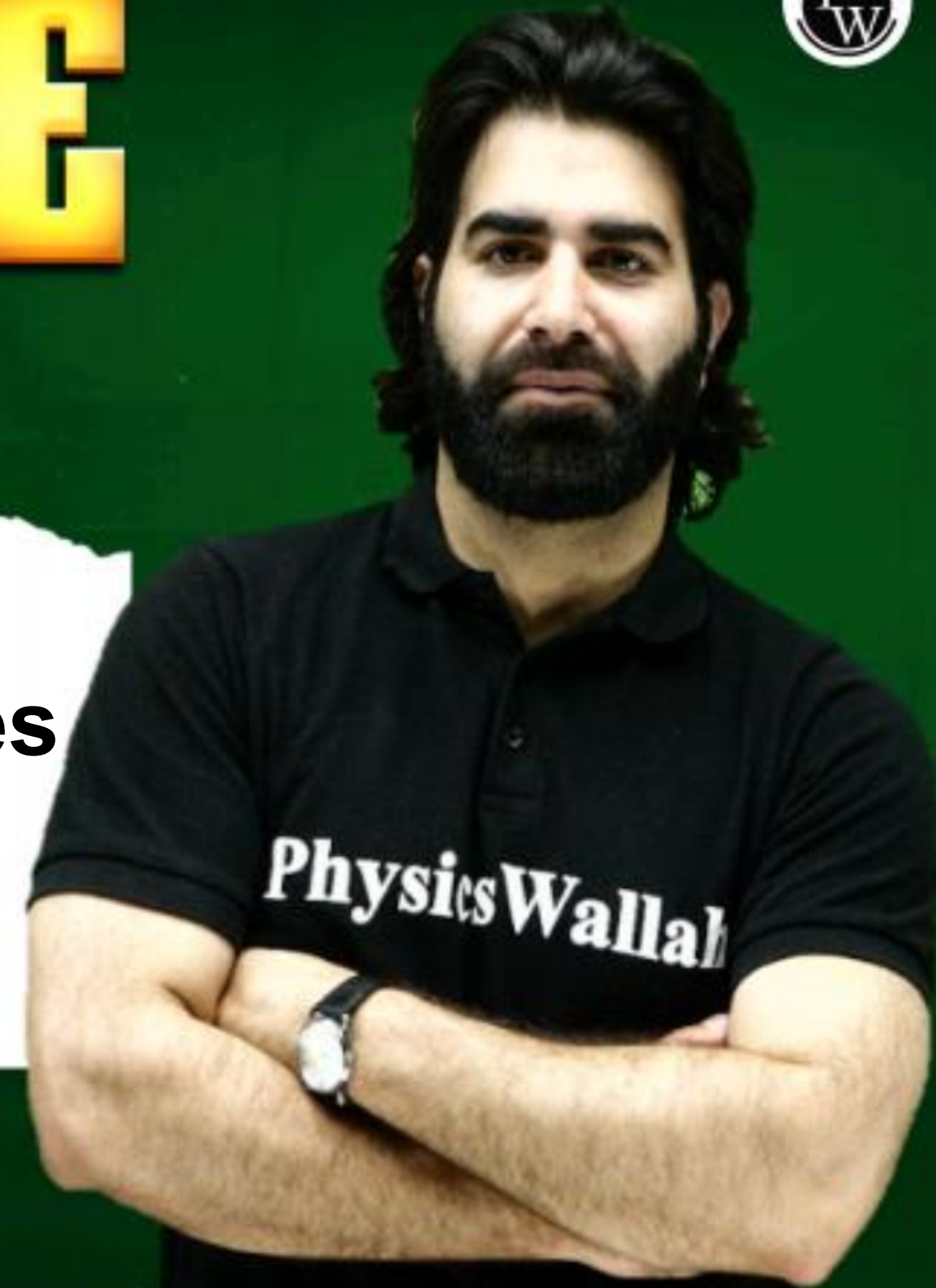
2026 CLASS-11TH

Breathing and Exchange of Gases

LECTURE-6

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Transport of Carbon dioxide
- 2 Regulation, Disorders

Revision

Last Class

Transport of O_2 - ① Dissolved oxygen 3 %
② OxyHb 97 %

→ Favorable Condition $PO_2 \uparrow, PCO_2 \downarrow, H^+ \downarrow, Temp \downarrow, BPG \downarrow$

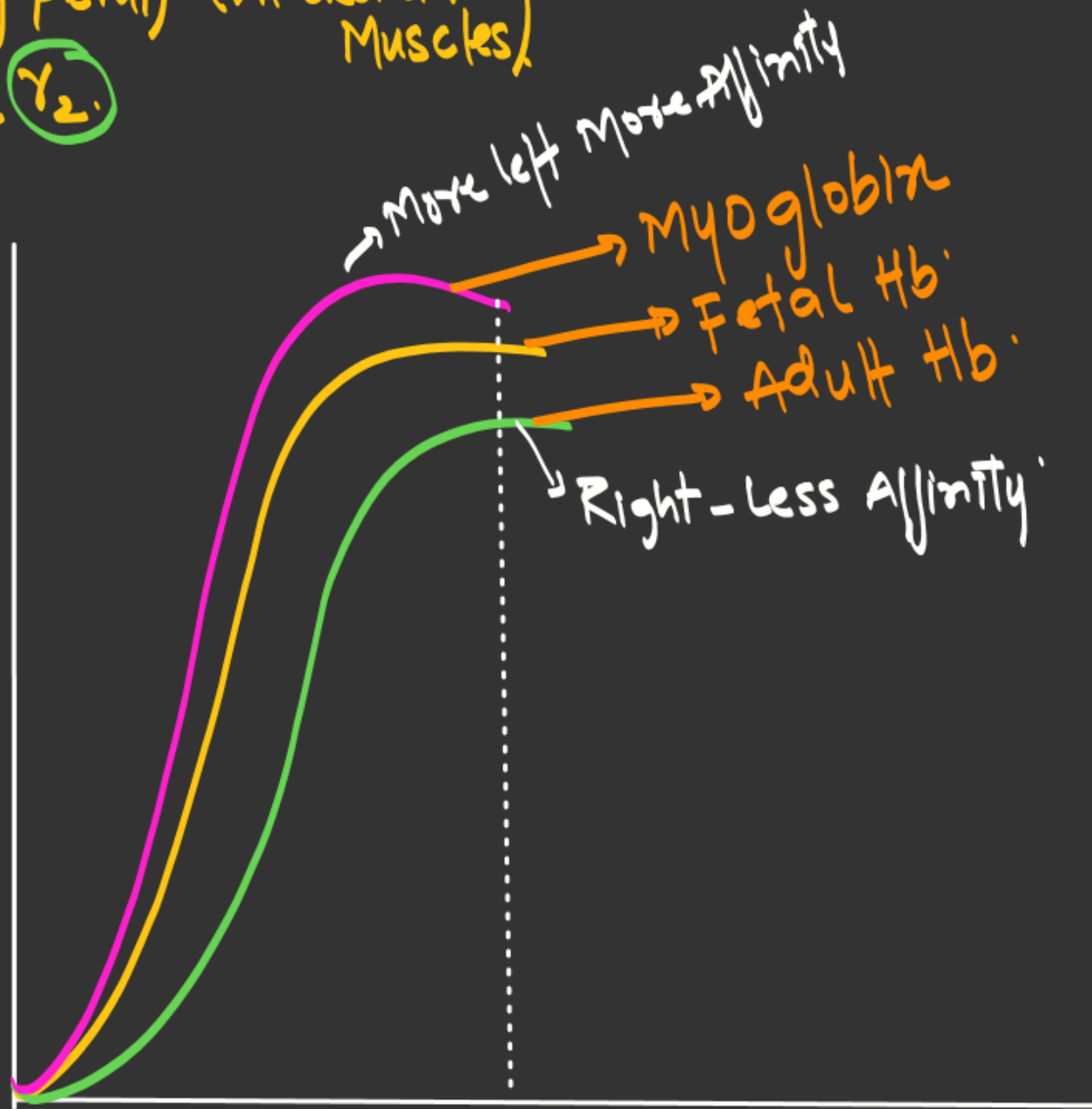
→ Unfavorable Condition $PO_2 \downarrow, PCO_2 \uparrow, H^+ \uparrow, Temp \uparrow, BPG \uparrow$

Oxy-Hb Curve Continue

→ Affinity of O_2 with Adult Hb, Fetal Hb, Myoglobin (Comparison).
- (After Birth) - (During Fetal) (In Skeletal Muscles)
- $\alpha_2\beta_2$ - $\alpha_2\gamma_2$

Order of Affinity

Myoglobin > Fetal Hb > Adult Hb.

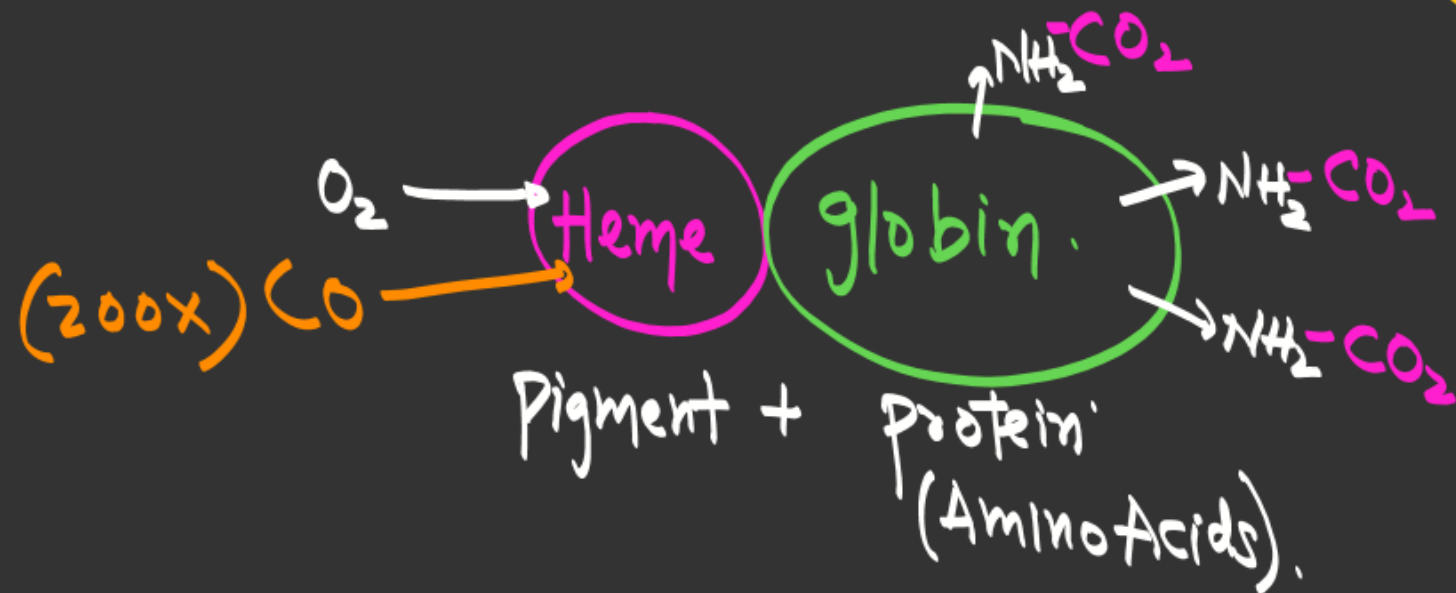


Transport of Carbon Dioxide.

- CO₂ is transported from Tissue to Lungs via 3 ways:→

✓ (a) as Dissolved CO₂: - 7%
- CO₂ is more soluble than Oxygen (20-25 times)

✓ (b) as Carbamino Haemoglobin: → 20-25%
- CO₂ Attaches to Amine Residues of Haemoglobin chain (Globin part).



Incomplete combustion

(CO Poisoning)

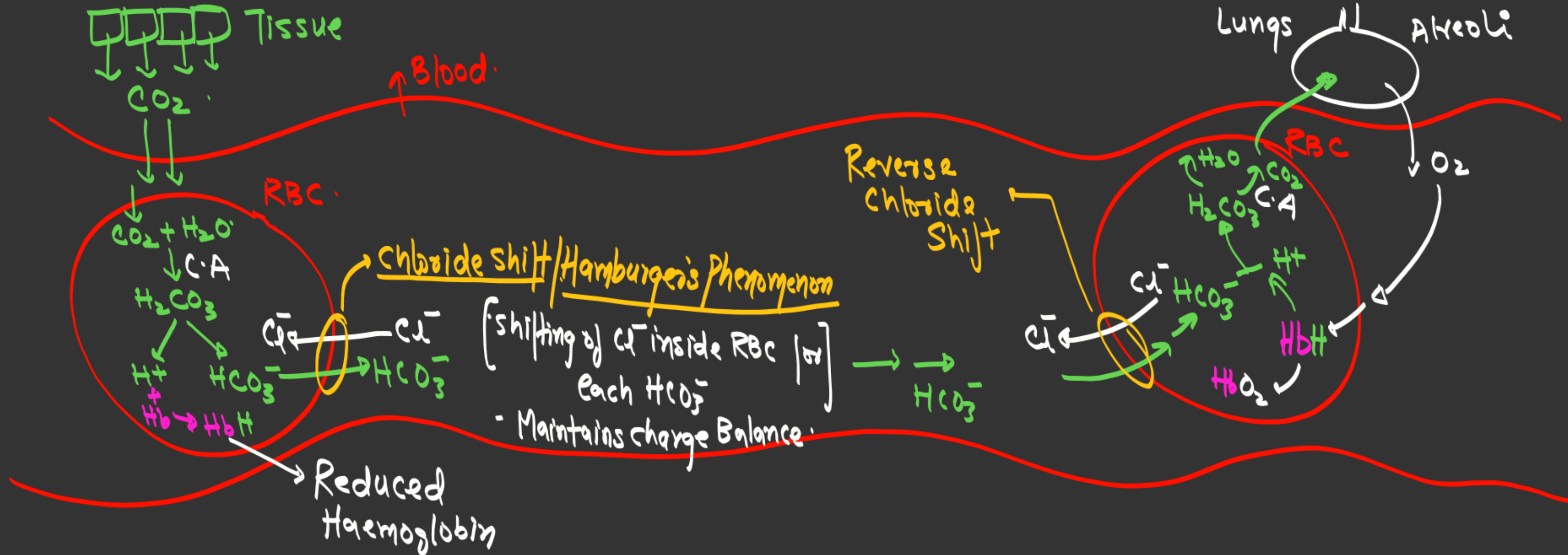
Carboxy Haemoglobin

- CO (carbon Monoxide) Binds to Heme of Hb instead of O₂.
- CO from Smoke.
- CO forms 200 times Stronger Bond

③ as Bicarbonate. 70%.

→ Majority of CO_2 is Transported as Bicarbonate (HCO_3^-).
→ $\text{CO}_2 + \text{H}_2\text{O} \xrightleftharpoons{\text{CA}} \text{H}_2\text{CO}_3 \longrightarrow \text{H}^+ + \text{HCO}_3^-$ → Another Function
Carbonic Acid. Acts as Buffer.

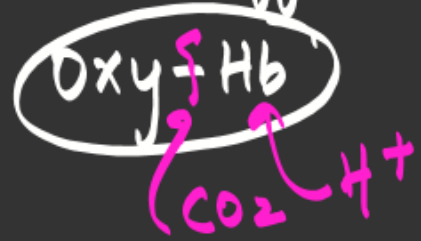
→ This Reaction Requires — Carbonic Anhydrase (In RBC).



Haldane Effect

- Movement of CO_2 from Blood into Lungs is due to formation of Oxy-Hb.
- Occurs at Lungs.

Bohr's Effect



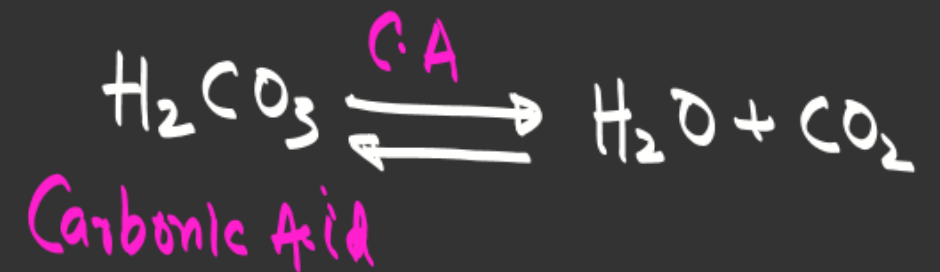
- Shifting of Oxy-Hb curve towards Right (Unfavourable) due to Increase in PCO_2 & H^+ Conc.
- Occurs at Tissues, causing release of O_2

Naam Karan.

- What gives name to

Carbonic Anhydrase

↓
Removes H_2O From Acid.



- Answer Because of the Backward Reaction

Regulation of Breathing

CPR.
→ Mouth to Mouth

Neural Regulation

It has centres in:

(a) Medulla.

- It has Respiratory Rhythm centre.

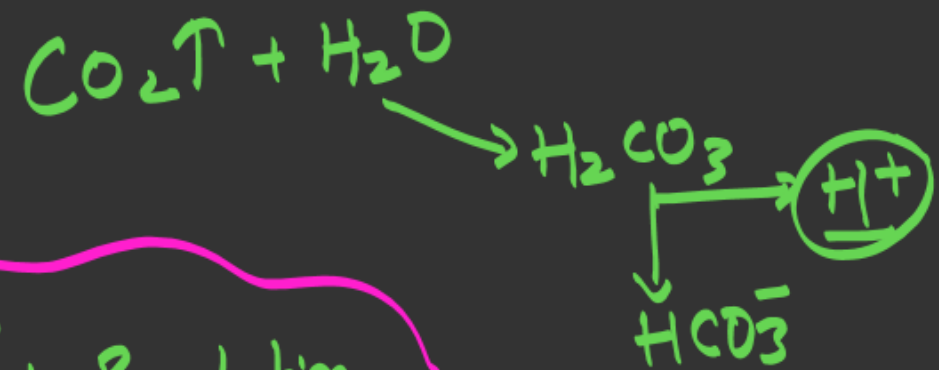
- It primarily Regulates Breathing.

→ It sends Message for Inspiration & Expiration.

(b) Pons.

- It has Pneumataxic centre.

- It can Moderate the Rhythm centre and changes Rate of Breathing.



Chemical Regulation:

→ It is having chemoreceptors.

→ These are sensitive to chemicals.

→ Sensitive to CO₂ & H⁺ More.

→ These Receptors are present in

(a) Brain → In chemosensitive Area.

- Near to Rhythm centre.

(b) Body → In Aorta, Carotid Artery.

- Rise in H⁺ & CO₂, Causes these to Send message to Rhythm centre to do Adjustments.

Asthma is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking.

H/W
Occupational Respiratory Disorders: In certain industries, especially those involving grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation. Long exposure can give rise to inflammation leading to fibrosis (proliferation of fibrous tissues) and thus causing serious lung damage. Workers in such industries should wear protective masks.

Disorders

① Emphysema.



- Damage to Alveoli
- Decrease in Surface area for gaseous exchange.
- Due to decreased Elasticity of Lungs.
- Often seen cigarette smokers.
- Chronic Disease.

↓
Time taken ↑

② Asthma.

- Allergic Disorder
- Inflammation of Bronchi & Bronchioles.
- Narrowing of Passage
- Difficulty in Breathing
- Causing wheezing.

QUESTIONS

The partial pressure of oxygen is equal in

- 1 Atmospheric air and Alveoli
- 2 Alveoli and Oxygenated blood
- 3 Alveoli and Deoxygenated blood
- 4 Deoxygenated blood and Tissues

QUESTIONS

Which of the following does not shift the oxy-haemoglobin dissociation curve to the right?

- 1 Increased pH
- 2 Increased carbon dioxide
- 3 Increased temperature
- 4 Increased 2,3 -dpg

Thank
You